

How to Switch from X11 to Wayland on Ubuntu and Debian

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The Linux display server world has been in transition for years now, and it can feel overwhelming trying to figure out if you should switch, when to switch, and honestly, how to even do it without breaking everything.

I've made this switch on multiple machines over the past few years, and while it's gotten significantly smoother, there are still some gotchas worth knowing about. Let me walk you through the entire process, from checking if you're ready to actually making the switch — and importantly, how to roll back if things don't go as planned.

Why Switch from X11 to Wayland Anyway?

Before we dive into the how, let's talk about the why for a second. X11 has served us well since the 1980s, but it's showing its age. Wayland was designed from the ground up to address many of X11's architectural limitations.

Here's what you actually gain with Wayland:

Better security. Wayland's architecture prevents applications from snooping on each other or injecting input events globally. With X11, any application can essentially spy on what you're typing in other windows. Not ideal in 2026.

Smoother graphics performance. Wayland eliminates a lot of the overhead that X11 carries around. You'll notice this especially if you're running on a laptop with a high DPI display or doing anything graphics-intensive.

Proper multi-monitor support. If you've ever tried running multiple monitors with different refresh rates on X11, you know the pain. Wayland handles this elegantly, with each monitor running at its own refresh rate without tearing or stuttering.

Better battery life on laptops. The improved efficiency isn't just theoretical — I've seen real-world improvements of 15–30 minutes of battery life on my ThinkPad after switching.

That said, Wayland isn't perfect yet. Some older applications don't play nice with it, and if you rely heavily on X11-specific tools like certain screen sharing applications or automation scripts using `xdotool`, you might want to think twice before switching.

Before You Switch: Compatibility Check

This is the part people usually skip, and then they wonder why their favorite screen recording tool suddenly doesn't work. Let's do this right.

Check Your Desktop Environment

First things first: not all desktop environments have equal Wayland support. Here's where things stand:

GNOME has the best Wayland support by far. It's been the default on Fedora since 2016 and has been Ubuntu's default since 21.04. If you're running GNOME, you're in good shape.

KDE Plasma has made huge strides. Plasma 5.25 and later work really well on Wayland, though you might still hit the occasional quirk with certain applications.

Other DEs vary widely. XFCE and MATE are still primarily X11-focused, though they're working on it. Sway is a Wayland-native i3-compatible tiling window manager if you're into that workflow.

Check which desktop environment you're running:

```
echo $XDG_CURRENT_DESKTOP
```

Check Your Graphics Drivers

Wayland works best with open-source graphics drivers. Here's the reality:

Intel graphics work brilliantly with Wayland. No issues here.

AMD graphics with the open-source AMDGPU drivers are excellent on Wayland. Again, smooth sailing.

NVIDIA is where it gets complicated. The proprietary NVIDIA drivers historically had terrible Wayland support, though version 495 and later have improved things significantly. If you're on NVIDIA, check your driver version:

```
nvidia-smi
```

You want version 495 or newer, and honestly, 515+ is where things really start working well. If you're on older drivers, you might want to update first or stick with X11 a bit longer.

Check Your Critical Applications

Some applications still don't work properly on Wayland, or work with limitations:

- **Screen sharing** works in most modern browsers and apps now, but some older tools still struggle
- **OBS Studio** works but needs specific Wayland support enabled
- **X11 automation tools** like xdotool won't work (there are Wayland alternatives like ydotool)
- **Some proprietary apps** like certain VPN clients might have issues

Make a list of your must-have applications and do a quick search to see if there are known Wayland issues. Trust me, it's better to know now than after you've switched.

How to Switch from X11 to Wayland on Ubuntu

Alright, assuming you've done your homework and you're ready to switch, here's how to do it on Ubuntu. The good news? If you're running Ubuntu 21.04 or later with GNOME, you probably already have everything you need installed.

Step 1: Ensure Wayland Session is Installed

On Ubuntu with GNOME, this should already be there, but let's verify:

```
dpkg -l | grep wayland
```

You should see packages like `libwayland-client0` and `libwayland-server0`. If you're on GNOME and don't see `gnome-shell-wayland`, install it:

```
sudo apt update
sudo apt install gnome-session-wayland
```

For KDE Plasma users:

```
sudo apt install plasma-workspace-wayland
```

Step 2: Log Out and Select Wayland Session

This is where the actual switch happens, and it's refreshingly simple:

1. Log out of your current session (don't reboot, just log out)
2. At the login screen, click on your username but don't enter your password yet
3. Look for a gear icon or settings icon in the bottom-right corner of the screen
4. Click it and you should see options like "Ubuntu" or "Ubuntu on Wayland" (or "Plasma (Wayland)" for KDE)
5. Select the Wayland option
6. Now enter your password and log in

That's it. You're now running Wayland.

Step 3: Verify You're Running Wayland

Once you're logged in, let's make sure it worked. Open a terminal and run:

```
echo $XDG_SESSION_TYPE
```

If this returns wayland, congratulations! You're running Wayland. If it says x11, something went wrong, and we'll cover troubleshooting in a bit.

You can also use this command for more detailed information:

```
loginctl show-session $(loginctl | grep $(whoami) | awk '{print $1}') -p Type
```

How to Switch from X11 to Wayland on Debian

The process on Debian is very similar to Ubuntu, with a few small differences depending on which version you're running.

For Debian 11 (Bullseye) and Later

Debian 11 and 12 come with Wayland support built-in for GNOME. The process is identical to Ubuntu:

```
# Verify Wayland packages are installed
dpkg -l | grep wayland

# Install GNOME Wayland session if needed
sudo apt update
sudo apt install gnome-session-wayland
```

Then log out and select the Wayland session at the login screen exactly as described for Ubuntu.

For Debian 10 (Buster) and Older

If you're still on Debian 10, Wayland support exists but isn't as polished. You might need to install additional packages:

```
sudo apt update
sudo apt install wayland-protocols libwayland-client0 libwayland-cursor0
libwayland-server0
```

Honestly though, if you're on Debian 10, I'd recommend upgrading to Debian 11 or 12 first. You'll get a much better Wayland experience, plus all the other improvements that come with newer releases.

Making Wayland Your Default Session

If everything works well and you want Wayland to be your default without having to select it every time you log in, you can set that up.

For GNOME on Ubuntu/Debian, edit the custom configuration:

```
sudo nano /etc/gdm3/custom.conf
```

Find the line that says:

```
#WaylandEnable=false
```

Make sure it's either commented out (with the #) or changed to:

```
WaylandEnable=true
```

Save the file and reboot. GDM3 will now prefer Wayland sessions by default.

For KDE users, the session you last used typically becomes the default automatically.

How to Switch Back to X11

Let's say Wayland isn't working out for you right now. Maybe an application you rely on doesn't work properly, or you're having stability issues. No problem — switching back to X11 is just as easy as switching to Wayland was.

Temporary Switch (Just This Session)

Log out, click the gear icon at the login screen, and select "Ubuntu" or "Plasma (X11)" instead of the Wayland variant. That's it.

Permanent Switch Back to X11

If you want to disable Wayland entirely and stick with X11:

Edit the GDM configuration:

```
sudo nano /etc/gdm3/custom.conf
```

Uncomment and change this line:

```
WaylandEnable=false
```

Save the file and reboot. Now even if you have the Wayland session installed, GDM won't offer it as an option.

Final Thoughts

Switching from X11 to Wayland on Ubuntu or Debian is pretty straightforward these days, especially if you're running a recent version with GNOME or KDE Plasma. The key is doing your homework first — check your desktop environment, verify your graphics drivers, and make sure your critical applications will work.

Start by just trying it out. Log in to a Wayland session and use it for a day. If something doesn't work right, you can always switch back to X11 with literally two clicks at the login screen. There's no risk to experimenting.